

SAT Practice Test 3 Answer Key

Complete Answers and Explanations

SECTION 1, MODULE 1: Reading and Writing

Question 1

Answer: D

(Explanation: In the passage, Tommy throws a stone “so far that it becomes a tiny dark speck before vanishing beneath the grey waves.” The word “hurl” means to throw something with great force, which matches the description of the stone traveling a great distance. “Strengthen” (A), “celebrate” (C), and “shape” (B) do not describe the physical act of propelling an object.)

Question 2

Answer: A

(Explanation: Lin’s memorial is described as reflecting the tradition of Japanese simplicity—anyone familiar with those gardens “would recognize the influence.” An “echo” is a reflection or reverberation of something, making it the best fit for a design that mirrors another tradition. A “dismissal” (B) implies rejection, a “critique” (C) implies criticism, and a “theory” (D) implies abstract speculation—none describe a design that pays tribute to an existing style.)

Question 3

Answer: A

(Explanation: The Singapore Exchange ensures transparency standards, which can reduce or ease investors’ concerns about regulatory gaps. “Alleviate” means to make something less severe or to relieve, fitting the context of reducing concerns. “Broadcast” (B) means to spread widely, “intensify” (C) means to strengthen, and “specify” (D) means to identify precisely—none logically describe what happens to concerns when transparency is ensured.)

Question 4

Answer: B

(Explanation: The passage says classifying musicians into rigid genres causes critics to miss subtleties. “Overlook” means to fail to notice, which directly matches “may inadvertently miss nuances” in the following sentence. “Condemn” (A) means to express disapproval, “exaggerate” (C) means to overstate, and “investigate” (D) means to examine—none capture the idea of unintentionally missing something.)

Question 5

Answer: C

(Explanation: The text shows Ifemelu guarding the survey “like a prized heirloom,” carrying it at all times, and approaching neighbors everywhere—at mailboxes, elevators, parking garages, and laundry rooms—regardless of what they are doing. This persistent, relentless behavior portrays her strong personal commitment to collecting survey responses. Choice A is unsupported because neighbors’ enthusiasm is not described. Choice B is wrong because no change over time is shown. Choice D is wrong because no refusals or personal dislike are mentioned.)

Question 6

Answer: A

(Explanation: The underlined sentence defines reduplication as a phenomenon “in which an element of a root word is repeated, sometimes with modification, within another word that is related to the root word.” This is a general definition of the linguistic concept. Choice B is wrong because the language’s geographic origin is mentioned in the first sentence, not the underlined one. Choice C is wrong because no contrast with other languages is made. Choice D is wrong because translations appear in the first sentence.)

Question 7

Answer: C

(Explanation: The text explicitly states that emulsifiers “are released into the water from nanoplastics and have a much stronger flow-reducing effect.” This directly answers the question. Choices A and B describe functions—containing or filtering nanoplastics—that are not mentioned in the passage. Choice D (composed of dissolved minerals) has no basis in the text.)

Question 8**Answer: A**

(Explanation: The text states the technology “is not yet commercially feasible as a large-scale power source” and describes ongoing engineering efforts to address challenges in storage, transport, distribution, cost, and efficiency. This supports the main idea that continued R&D is needed before widespread use. Choice B incorrectly states the purpose was long-term commercial storage. Choice C incorrectly claims a recent increase in commercial use. Choice D makes an unsupported efficiency comparison with plants.)

Question 9**Answer: C**

(Explanation: The table shows Nisga’a Lands in British Columbia has an area of 2,019 square kilometers. The student wants to compare his territory (Tłıchǫ Lands, 39,000 sq km in the Northwest Territories) with territories in other provinces. Nisga’a Lands is in British Columbia (a different province), so 2,019 sq km completes the comparison. Choice D (41,437) is the Sahtu Settlement Area—also in the Northwest Territories, not a different province as required.)

Question 10**Answer: D**

(Explanation: Italy reported 87 insect species, Portugal reported 22, and Greece reported 7. Italy’s count (87) is indeed greater than both Portugal (22) and Greece (7), directly supporting the researchers’ conclusion. Choice A discusses fungus-insect comparisons within countries but doesn’t directly address the conclusion. Choice B compares fungus counts, not insects. Choice C compares shrub counts, which is irrelevant to a conclusion about damaging insects.)

Question 11**Answer: A**

(Explanation: The conclusion states that Chen’s data would need less adjustment to fit the database classifications than Schmidt’s data. A higher classification alignment rating directly supports this because, as defined in the passage, a higher alignment rating means the study’s compound categories more closely match the database’s categories—thus requiring less adjustment. Choice C

about specificity is not directly relevant to classification adjustment. Choices B and D provide isolated ratings without the comparative element the conclusion requires.)

Question 12

Answer: D

(Explanation: Local residents ranked the supply of fresh vegetables at 3 out of 20, while the other listed benefits received rankings of 10, 18, 14, and 16. Since a lower rank number indicates a higher perceived contribution, rank 3 is the highest among all five listed benefits for local residents. This supports the official's conclusion that ads for residents should emphasize fresh vegetables. Choice A misstates the ranking relationship. Choice B compares volunteers and residents without addressing the conclusion. Choice C discusses a different benefit entirely.)

Question 13

Answer: A

(Explanation: If the light signals are directed at predators rather than used for intraspecies communication, we would expect the light to be triggered specifically by predator encounters and to cease once the threat passes. Choice A shows exactly this: no species produced light before encounters, and nearly all stopped within ten seconds after. This pattern strongly supports an anti-predator function. Choice B actually weakens the claim by showing krill flashes aren't detectable by predators. Choices C and D describe signal variety and species-specificity, which don't directly address whether signals target predators.)

Question 14

Answer: D

(Explanation: The passage shows that Hindi-speaking immigrants from northern India settled in Southall, while Sylheti/Bengali-speaking immigrants from the Sylhet region of Bangladesh settled in Tower Hamlets. This parallel between geographic origin and language on one hand, and London settlement patterns on the other, is best explained by choice D: geographic differences in primary language within South Asia are reflected in settlement patterns. Choice A introduces an unsupported claim about multilingualism. Choice B makes an unsupported comparison about total speaker numbers.

Choice C contradicts the text by suggesting frequent moves between boroughs.)

Question 15

Answer: D

(Explanation: “Making the pioneering nurse one of many healthcare professionals to appear on public health documents” is grammatically correct. The infinitive “to appear” functions as a modifier of “professionals,” describing what they did. Choice A (“did contribute”) creates an awkward past tense construction. Choice B (“had contributed”) introduces an unnecessary and incorrect past perfect. Choice C (“contributes”) creates a subject-verb agreement error with the plural “professionals.”)

Question 16

Answer: C

(Explanation: A nonrestrictive clause like “which is 6 miles long” provides additional information and must be set off by commas on both sides. Choice C correctly places commas before “which” and after “long,” properly enclosing the nonrestrictive clause within the sentence. Choice A omits the closing comma. Choice B uses a dash instead of a comma and omits the opening comma. Choice D incorrectly uses a colon.)

Question 17

Answer: A

(Explanation: “Montana, spanning 4,506 feet in elevation, reaches its highest point at Granite Peak” requires a participial phrase. The main verb of the sentence is “reaches.” The participle “spanning” correctly modifies “Montana” without creating a run-on or comma splice. “Spans” (B) would create two competing main verbs. “Span” (C) is a bare infinitive that doesn’t fit. “Spanned” (D) creates a past tense that conflicts with the present-tense “reaches.”)

Question 18

Answer: B

(Explanation: The dash before “Norway” correctly introduces the appositive that identifies the specific IMF member nation. At the end of

a sentence, a single dash is appropriate for this emphatic identification. Choice A (comma) is weaker but acceptable in other contexts. Choice C omits all punctuation, running the words together. Choice D (colon) is less conventional when the appositive is a single proper noun at sentence end.)

Question 19

Answer: C

(Explanation: “Conservation efforts that, through habitat restoration and scientific research, aim to bring tigers back...” The relative clause “that...aim” is interrupted by the parenthetical phrase “through habitat restoration and scientific research,” which must be enclosed in commas. Choice C provides the first necessary comma after “that.” Choices A (dash), B (semicolon), and D (colon) do not correctly punctuate a parenthetical interruption within a relative clause.)

Question 20

Answer: A

(Explanation: The subject is compound: “The star graph and the cycle graph.” A compound subject joined by “and” takes a plural verb, so “are” is correct. “Has been” (B), “is” (C), and “was” (D) are all singular forms that do not agree with the compound plural subject.)

Question 21

Answer: C

(Explanation: The phrase “claims dendrochronologist Valerie Trouet” is an inverted attribution (like “says Dr. Smith”). In this construction, no punctuation is needed between the job title and the name—they form a single noun phrase identifying one individual. Adding a comma (A), dash (B), or colon (D) would incorrectly separate the title from the name.)

Question 22

Answer: B

(Explanation: The Alvin can reach depths exceeding 6,000 meters, and most hydrothermal vents are found at extreme depths. As a result of this capability, the Alvin can provide more data about these

vents. “Consequently” correctly signals this cause-and-effect relationship. “In contrast” (A) signals opposition. “Additionally” (C) signals a mere addition without the causal link. “Actually” (D) signals a correction.)

Question 23

Answer: A

(Explanation: The passage notes that confetti was historically made from ledger rolls, then says these rolls had been replaced by computers by 2012. “Of course” signals that this transition to computers is an obvious, expected development the reader would naturally anticipate. “For example” (B) would introduce an illustration. “In addition” (C) would add a new point. “As a result” (D) would incorrectly imply the replacement was caused by confetti production.)

Question 24

Answer: A

(Explanation: The goal is to emphasize a difference between the two geckos. Choice A directly contrasts a specific physical feature (the tails) between the two species: the crested gecko’s is long and prehensile, whereas the mourning gecko’s is not described that way. Choices B and C describe only one gecko each without comparison. Choice D merely lists both geckos among Puerto Rico’s species without highlighting any difference.)

Question 25

Answer: B

(Explanation: The goal is to emphasize a difference. Choice B clearly identifies how the calendars differ in their leap year rules, providing specific details about the Julian approach (every 4 years without exception) versus the Gregorian modification (additional rules for accuracy). Choice A describes historical facts without emphasizing a specific difference. Choices C and D describe only one calendar each.)

Question 26

Answer: A

(Explanation: The goal is to compare the intensity of three wind categories. Choice A systematically compares all three by their Beaufort numbers, establishing a clear ordering: storm (10) > near gale (7) >

gentle breeze (3). Choice B merely classifies the categories without ranking them. Choice C compares only two categories and omits the near gale. Choice D discusses the scale's maritime origins, which is irrelevant to comparing intensity.)

Question 27

Answer: B

(Explanation: The goal is to begin a narration of Huerta's co-founding of the National Farmworkers Association. Choice B uses vivid, narrative-style language ("It was in September 1962, in the agricultural town of Delano, California...") to set the scene with specific time, place, and purpose—an effective opening for a narration. Choice A presents a factual comparison. Choice C focuses on archives. Choice D summarizes events without narrative style.)

SECTION 1, MODULE 2: Reading and Writing

Question 1

Answer: D

(Explanation: Transportation costs could cancel out or eliminate the financial benefits of remittances for families in remote villages. "Nullify" means to make something void or ineffective, precisely describing how steep costs could negate the benefits. "Underestimate" (A) means to judge as less than actual, "misconstrue" (B) means to interpret wrongly, and "denigrate" (C) means to criticize unfairly—none describe the cancellation of a benefit.)

Question 2

Answer: C

(Explanation: The art installations require software that is "largely unsupported since 2017" (specifically Java applets). "Defunct" means no longer existing or functioning, which accurately describes software that is no longer supported and cannot run in modern browsers. "Extraneous" (A) means irrelevant or unnecessary, "arcane" (B) means mysterious or obscure, and "ubiquitous" (D) means found everywhere—none capture the idea that the software has ceased to function.)

Question 3**Answer: D**

(Explanation: Lowering the tax threshold could hinder economic growth by “increasing the administrative burden” during a critical scaling period. “Stymie” means to prevent or hinder progress, which matches the idea of administrative burdens obstructing entrepreneurial growth. “Disparage” (A) means to belittle, “outstrip” (B) means to surpass, and “compound” (C) means to make worse or to combine—none precisely describe the hindering of growth.)

Question 4**Answer: B**

(Explanation: The surge in vinyl record values had “the counterintuitive effect of _____ demand”—new collectors flooded the market. “Engendering” means causing or giving rise to something. The price surge unexpectedly engendered new demand by attracting new collectors. “Exploiting” (A) means taking advantage of, “monetizing” (C) means converting to revenue, and “appraising” (D) means assessing value—none describe the creation of new demand.)

Question 5**Answer: C**

(Explanation: The text first presents Yamamoto’s study concluding that ocean temperatures affect tuna migration, then introduces Mendez’s concern that the small sample size (30 fish) could lead to biased analysis and exaggerated effects. This two-part structure—result followed by concern—matches choice C. Choice A is wrong because no similarity between two studies is stated. Choice B is wrong because the text discusses migration patterns, not species characteristics. Choice D is wrong because no solution is proposed.)

Question 6**Answer: B**

(Explanation: The underlined sentence says the model “is sure to be invaluable in determining the optimal mix of a district’s businesses.” This directly emphasizes the potential usefulness (utility) of the research team’s predictive model. Choice A overstates—the sentence praises the model, not specifically the algorithm’s accuracy. Choice C is wrong because no trends in San Francisco are summarized. Choice

D describes the sentence that precedes the underlined one, not the underlined sentence itself.)

Question 7

Answer: D

(Explanation: The text describes prior work by Alvarez (shared pathways) and Sharma (different pathways), notes a gap in knowledge (relative frequency is “still poorly understood”), then explains that Tanaka and Okonkwo were motivated to address this gap in their 2019 paper. This structure situates their work within the broader field of evolutionary biology research. Choice A wrongly suggests their conclusions need strengthening. Choice B mischaracterizes their contribution. Choice C is wrong because no criticism of Alvarez is presented.)

Question 8

Answer: C

(Explanation: Text 1 argues that scholarship overemphasizes confessional modes while ignoring works like Spark’s satirical fiction. Text 2 argues that scholars have focused too narrowly on “confessional sincerity versus ironic detachment,” missing other stylistic dimensions. Both authors agree that current scholarship only partially captures the full stylistic picture of postwar fiction. Choice A is unsupported—neither discusses obsolete methods. Choice B mischaracterizes both texts’ goals. Choice D is wrong because both authors advocate widening the focus on style, not moving beyond it.)

Question 9

Answer: A

(Explanation: Lindqvist et al. propose that community solidarity motivates lower-income voters to support programs funded by taxes on themselves. The key insight is that social bonds (strong community ties) influence voting behavior, causing people to take communal welfare into account when voting. Choice B incorrectly states values coincide with material interests—the point is they override them. Choice C focuses on politicians’ strategies, not voter motivation. Choice D incorrectly frames the relationship as a conflict rather than a motivation.)

Question 10**Answer: B**

(Explanation: The maximum density reported by Liberg (Scandinavia) was 2.10, compared to Ciucci's (Abruzzo) maximum of 7.80. Since population density reflects how many wolves a habitat can support per 1,000 km², this substantial difference suggests Scandinavia's habitat cannot support as many wolves as Abruzzo can. Choice A makes a trivial claim about Tuscany. Choice C draws an unsupported conclusion about land area proportions. Choice D compares total wolf populations, which the table's density data cannot determine.)

Question 11**Answer: C**

(Explanation: The claim is that Lily "admires wealth in other people to the exclusion of all other qualities." Choice C shows this perfectly: "Lily was immensely interested in people who possessed great fortunes. Otherwise not at all." The phrase "otherwise not at all" makes clear that wealth was the sole quality that captured her interest—no wealth, no interest. Choices A and B mention luxury and ambition but don't establish the exclusion of all other qualities. Choice D describes her appearance, not her admiration of wealth.)

Question 12**Answer: C**

(Explanation: The hypothesis is that HVAC filter accumulation could serve as a VOC detection method, eliminating the need for costly PIDs. If filters can detect even low (harmless) concentrations more easily than PIDs detect high (harmful) ones, this strongly supports the filters' usefulness as a replacement monitoring tool—they would be more sensitive and practical. Choice A (comparable rates) merely shows parity, not superiority. Choice B discusses filter durability, not detection ability. Choice D suggests inconsistency in filter readings, which would undermine the hypothesis.)

Question 13**Answer: C**

(Explanation: The conclusion is that younger cats are worse at differentiating familiar from unfamiliar scents. Choice C directly supports this: cats showed different brain activity patterns for familiar

vs. unfamiliar scents, and this difference decreased as the age of the cat decreased (i.e., younger cats). Smaller differences in brain activity between the two scent types means less effective differentiation. Choice A describes declining overall activity, not differentiation ability. Choice B describes a pattern between scrambled and unfamiliar scents. Choice D contradicts the conclusion by suggesting sensitivity declines with age.)

Question 14

Answer: D

(Explanation: The claim is that new freedoms and possibilities have become available to the working class. Choice D describes how workplaces (“mill and furnace, forge and field”) have “opened wider paths before the common hand” with “greater yields” that are “more justly earned”—directly illustrating expanded opportunities and freedoms for workers. Choice A describes simple labor without new freedoms. Choice B describes burdens and toil, the opposite of new possibilities. Choice C references horizons but credits hardship, not newly available freedoms.)

Question 15

Answer: D

(Explanation: The passage states that conditions during neutron star formation are believed to have been “ill-suited to producing such extreme spin rates.” If some pulsars still spin extremely fast despite these unfavorable conditions, it suggests that our current understanding of the formation or evolution process may be incomplete—something is missing from the model. Choices A, B, and C make specific claims about individual pulsars or age comparisons that are not supported by the passage.)

Question 16

Answer: C

(Explanation: The hypothesis says chromatophore displays evolved because they have lower metabolic cost than ink defense. But if species with chromatophore displays also produce highly toxic ink, they clearly haven’t abandoned the ink defense in favor of the less costly alternative. This suggests the metabolic-cost hypothesis doesn’t fully explain why all species developed chromatophore displays—other factors must be involved. Choices A and B overstate

the implications. Choice D introduces a distinction between the hypothesis and findings that is not clearly articulated.)

Question 17

Answer: C

(Explanation: A colon after “initiative” correctly introduces the specific name and description of the initiative. “Timeline of Soul Food” is an appositive that elaborates on what the “important new initiative” is, and a colon is the standard punctuation for introducing such an identification. Choice A lacks any punctuation, creating a run-on. Choice B uses a semicolon, which would require two independent clauses. Choice D creates two separate sentences, breaking the grammatical connection.)

Question 18

Answer: D

(Explanation: When a title is followed by a specific name identifying one individual, no punctuation is needed: “linguist Nomso Dlamini of the University of Cape Town.” The title and name form a single unit (a restrictive appositive). Choice A (comma after “linguist”) incorrectly separates the title from the name. Choice B (colon) incorrectly implies a list or explanation follows. Choice C adds commas that would make the name nonrestrictive, implying there is only one linguist in the world.)

Question 19

Answer: B

(Explanation: A colon after “centuries” introduces the name and description of the mercantile alliance. The text that follows (“the Hanseatic League, a loose confederation...”) identifies what the alliance was, and a colon is standard for introducing such an identification. Choice A (semicolon) would require what follows to be an independent clause. Choice C omits necessary punctuation. Choice D (period) breaks the sentence into two, losing the grammatical relationship.)

Question 20

Answer: D

(Explanation: The subject is “collection” (singular), not “artworks” or “artists.” “Whose extensive collection...ranks among the finest” uses the singular verb “ranks” to agree with the singular subject “collection.” The intervening prepositional phrase “of artworks by European masters and contemporary artists” does not change the subject. Choices A (“have been ranked”), B (“rank”), and C (“are ranked”) are all plural forms that do not agree with the singular “collection.”)

Question 21

Answer: B

(Explanation: The names “Scarface and Luna” are essential (restrictive) appositives that identify which specific sharks. No commas are needed because the names are necessary information—without them, we wouldn’t know which sharks are meant. Choice A adds an unnecessary comma after “Luna.” Choice C adds commas around the names, making them nonrestrictive, which would incorrectly imply that “the sharks” had already been identified. Choice D adds a comma before the names but not after, creating an inconsistency.)

Question 22

Answer: D

(Explanation: The sentence needs “Macharia” as the subject of the main verb “celebrates.” The participial phrase “having referred to Kenya’s thriving film scene...” is set off by a comma as a nonrestrictive modifier. “Macharia, having referred..., celebrates” is grammatically correct—the participial phrase provides background information while “celebrates” serves as the main verb. Choice A (“has”) creates a comma splice with the second clause. Choice C places a comma between subject and verb. Choice B lacks a necessary comma before the participial phrase.)

Question 23

Answer: A

(Explanation: The passage compares an older solar park (40 MW, 2008) with a newer one (187 MW, 2020). “In contrast” correctly signals this comparison between two different performance levels. “Granted” (B) concedes a point. “Eventually” (C) suggests a temporal progression that isn’t the focus. “For this reason” (D) implies a causal relationship that doesn’t exist between the two parks.)

Question 24**Answer: C**

(Explanation: The goal is to specify Titanoboa's location. Choice C directly states where a reconstructed model can be found: the Smithsonian National Museum of Natural History in Washington, D.C. This combines the specimen name with its precise location. Choice A leads with the city rather than Titanoboa. Choice B discusses when Titanoboa lived, not where to find it. Choice D describes taxonomy, not location.)

Question 25**Answer: B**

(Explanation: The goal is to compare the acidity or basicity of three substances. Choice B provides specific pH values for all three substances, classifies each as acidic or basic, and ranks them relative to each other (lemon juice most acidic, bleach most basic). Choice A mentions all three fall at different points but doesn't specify their relative positions. Choices C and D each discuss only one substance.)

Question 26**Answer: A**

(Explanation: The goal is to begin a narrative about Carmen Miranda's life. Choice A establishes the setting (Marco de Canaveses, Portugal) and the time (1909) in a narrative style ("her story begins"), making it an effective chronological opening for a biography. Choice B crams too many facts together without narrative flow. Choice C starts in 1943, midway through her career. Choice D begins with her career, not her life's beginning.)

Question 27**Answer: B**

(Explanation: The goal is to connect the McAllester quotation to a specific album. Choice B directly links the quotation (about drawing on Native flute traditions) to the specific album Changes (which incorporates a traditional cedar flute), effectively connecting the critical observation to a concrete work. Choice A restates the quotation without naming an album. Choice C mentions the album but drops the quotation. Choice D paraphrases without naming the album.)

SECTION 2, MODULE 1: Math

Question 1

Answer: 36

(Explanation: The 144° angle and angle y are co-interior (same-side interior) angles, so they sum to 180° . Thus $y = 180 - 144 = 36$.)

Question 2

Answer: D

Question 3

Answer: D

(Explanation: $56x = 8(7x) = 8(3) = 24$.)

Question 4

Answer: A

(Explanation: Distribute the 3: $x^3 + 6x^2 - 5x + 3x^2 + 36 = x^3 + 9x^2 - 5x + 36$.)

Question 5

Answer: 32

(Explanation: $f(3) = 3^3 + 5 = 27 + 5 = 32$.)

Question 6

Answer: 1272

(Explanation: $x/4 = 318$, so $x = 318 \times 4 = 1272$.)

Question 7

Answer: D

(Explanation: The data shows a negative linear trend with a y-intercept around 9–10 and a negative slope of roughly -0.9 . Option D fits best.)

Question 8**Answer: A**

(Explanation: The vertex is at (6, 3). Since the coefficient $1/16 > 0$, the parabola opens upward, so the vertex represents the minimum. The minimum height is 3 feet.)

Question 9**Answer: A**

(Explanation: Check $x=0$: $5(0) - 3(15) = -45$ ✓. Check $x=3$: $5(3) - 3(20) = 15 - 60 = -45$ ✓. Check $x=6$: $5(6) - 3(25) = 30 - 75 = -45$ ✓.)

Question 10**Answer: 45****Question 11****Answer: B**

(Explanation: Set equal: $-(1/5)x = (1/13)x \rightarrow -(1/5)x - (1/13)x = 0 \rightarrow x(-1/5 - 1/13) = 0 \rightarrow x = 0$.)

Question 12**Answer: D****Question 13****Answer: 40**

(Explanation: The population doubles when $2^{(t/40)} = 2$, which occurs when $t/40 = 1$, so $t = 40$.)

Question 14**Answer: C**

(Explanation: For infinitely many solutions, the two equations must represent the same line. Therefore, the slope of the second equation must equal the slope of the first: $3/8$.)

Question 15**Answer: D**

(Explanation: Divide the equation by 6: $x^2 - 3x - 6 = 0$, so $x^2 - 3x = 6$.)

Question 16**Answer: 864**

(Explanation: Side = $\sqrt[3]{1728} = 12$. Surface area = $6 \times 12^2 = 6 \times 144 = 864$.)

Question 17**Answer: 18**

(Explanation: Before coupon: $\$91 + \$35 = \$126$. Per notebook: $\$126 \div 7 = \18 .)

Question 18**Answer: 15**

(Explanation: For no real solutions, discriminant < 0 : $b^2 - 4(-3)(-20) < 0 \rightarrow b^2 - 240 < 0 \rightarrow b^2 < 240 \rightarrow b < \sqrt{240} \approx 15.49$. Largest positive integer is 15.)

Question 19**Answer: C**

(Explanation: In right triangle DEF with right angle at F, $\tan E = \text{opposite/adjacent} = DF/EF$. So $1/5 = 18.6/EF \rightarrow EF = 18.6 \times 5 = 93.0$.)

Question 20**Answer: B**

(Explanation: The minimum of $(x - 5)^2$ is 0, occurring at $x = 5$. Minimum of $f(x) = 0 + 9 = 9$.)

Question 21**Answer: D**

(Explanation: A is at $(0,1)$ and C is at $(0,-1)$. The angle ABC from ray BA to ray BC is π radians. Angles equivalent to π : $\pi + 2k\pi$. $21\pi = \pi + 20\pi = \pi + 10(2\pi)$. So 21π is equivalent to π .)

Question 22

Answer: D

(Explanation: We know $\angle J = \angle M = 45^\circ$. We know $KL = 20$ and $NO = 15$, giving a ratio of $4/3$. For SAS similarity, we'd need the sides forming the known angle to be proportional. KL and NO are opposite the known angles, not adjacent to them. Statement I gives $JK/MN = 4/3$ (one side adjacent to the angle), and Statement II gives $JL/MO = 4/3$ (another side adjacent). Either one alone with the angle would give only one proportional side pair adjacent to the angle plus one opposite — not sufficient for SAS. We'd need BOTH I and II together for SAS, but each alone is not sufficient. Since the question asks about each independently, neither alone is sufficient.)

SECTION 2, MODULE 2: Math

Question 1

Answer: 13

(Explanation: $f(0) = 6(0) + c = c = 13$.)

Question 2

Answer: C

(Explanation: Median of P (6 values) $= (10 + 10)/2 = 10$. Median of Q (7 values) $= \text{middle value} = 10$. Both medians are 10.)

Question 3

Answer: A

(Explanation: The confidence interval is 0.35 ± 0.06 , which is $(0.29, 0.41)$. It is plausible the true proportion lies within this range.)

Question 4

Answer: B

(Explanation: The shaded region is above/left of a line with positive slope passing through the origin. Point (0, 6) is above the line and in the shaded region.)

Question 5

Answer: A

(Explanation: Check $f(0) = 10$ for all: A) $0+0+10=10$ ✓, B) $0+0+10=10$ ✓, C) $0-0+10=10$ ✓, D) $0+0+10=10$ ✓. Check $f(1)$: A) $2+6+10=18$ ✓, B) $7+1+10=18$ ✓, C) $7-1+10=16$ ✗. Check $f(-1)$: A) $2-6+10=6$ ✓, B) $7-1+10=16$ ✗. Answer is A.)

Question 6

Answer: C

(Explanation: At $x = 0$, $y \approx 5$. At $x = 1$, $y \approx 7$. Ratio: $7/5 = 1.4$.)

Question 7

Answer: A

Question 8

Answer: B

(Explanation: In a 30-60-90 triangle, the side opposite 30° is the shortest. Here the 30° angle is at the bottom-right. The side opposite 30° is the vertical side w . The side adjacent to 30° is the horizontal side 48. $\tan(30^\circ) = w/48 \rightarrow w = 48 \times \tan(30^\circ) = 48 \times (1/\sqrt{3}) = 48/\sqrt{3} = 16\sqrt{3}$.)

Question 9

Answer: D

(Explanation: $q/r = 3150/4200 = 3/4$. When $r = 3360$: $q = (3/4)(3360) = 2520$.)

Question 10

Answer: 9600

(Explanation: Side $= \sqrt[3]{64000} = 40$. Surface area $= 6 \times 40^2 = 6 \times 1600 = 9600$.)

Question 11**Answer: D**

(Explanation: Distribute: $(6cx + 24)/5 = 3x - 2 \rightarrow 6cx + 24 = 15x - 10 \rightarrow (6c - 15)x = -34$. For no solution, the coefficient of x must be 0 and the constant nonzero: $6c - 15 = 0 \rightarrow c = 15/6 = 5/2$. And $-34 \neq 0$ ✓.)

Question 12**Answer: A**

(Explanation: $f(x) = 7x + 21$. The x -intercept is where $f(x) = 0$: $7x + 21 = 0 \rightarrow x = -3 = -21/7$. The x -intercept means when the output is 0, the input is $-21/7$.)

Question 13**Answer: B**

(Explanation: $3/c - 5/d = 4 \rightarrow 3/c - 4 = 5/d \rightarrow (3 - 4c)/c = 5/d \rightarrow d = 5c/(3 - 4c)$.)

Question 14**Answer: 3600**

(Explanation: Let a = votes against. Votes for = $6a$. Also, $6a - a = 18,000 \rightarrow 5a = 18,000 \rightarrow a = 3,600$.)

Question 15**Answer: D**

(Explanation: $f(12) = 3(12^2) = 3(144) = 432$.)

Question 16**Answer: 4/5 (or 0.8)**

(Explanation: $\sqrt[4]{p^3} = p^{3/4}$. So $p^{3/4} = t^{5/4} = (p^{(2n-1)})^{5/4} = p^{5(2n-1)/4}$. Setting exponents equal: $3/4 = 5(2n - 1)/4 \rightarrow 3 = 10n - 5 \rightarrow 10n = 8 \rightarrow n = 4/5$.)

Question 17**Answer: C**

(Explanation: $|x - 5| = k/6$. An absolute value equation has exactly one solution when $k/6 = 0$. If $k/6 > 0$, there are two solutions. If $k/6 < 0$, there are no solutions.)

Question 18**Answer: 18**

(Explanation: For no real solutions: $b^2 - 4(-5)(-18) < 0 \rightarrow b^2 - 360 < 0 \rightarrow b^2 < 360 \rightarrow b < \sqrt{360} \approx 18.97$. Largest positive integer b is 18.)

Question 19**Answer: D**

(Explanation: The y-intercept is at $x = 0$: $f(0) = a(1 + 2.5^b)$. For $g(x) = a(2.5^x + k)$: $g(0) = a(1 + k)$. The y-intercept value $a(1 + k)$ is not displayed as a single constant or coefficient. For $h(x) = a(2.5)^x + m$: $h(0) = a + m$. The y-intercept value $a + m$ is also not displayed as a single constant or coefficient. Neither form displays the y-coordinate of the y-intercept as a constant or coefficient.)

Question 20**Answer: D**

(Explanation: Cost = 180 for first hour + 95 for each additional hour ($x - 1$ additional hours). $y = 180 + 95(x - 1) = 180 + 95x - 95 = 95x + 85$.)

Question 21**Answer: 60000**

(Explanation: Let the mass of $Y = Y$. Mass of $X = 3.75Y$. Mass of $X = 0.00625Z \rightarrow Z = X/0.00625 = 3.75Y/0.00625 = 600Y$. So $Z = 600Y$, meaning Z is 60,000% of Y . Therefore $q = 60,000$.)

Question 22**Answer: D**

(Explanation: We know $\angle A = \angle D = 40^\circ$ and $BC/EF = 18/12 = 3/2$. BC and EF are opposite the known equal angles, not sides that include

the angles. For SAS similarity, we need two sides that include the known angle to be proportional. Statement I gives $AB/DE = 3/2$ (one side forming the angle), and Statement II gives $AC/DF = 3/2$ (the other side forming the angle). Either one alone gives us AA? No—we only have one pair of equal angles. With one pair of equal angles plus one pair of proportional sides adjacent to that angle, we cannot conclude similarity. We would need both I and II together (SAS), or another angle. So neither alone is sufficient.)

SECTION 1, HARD MODULE 2: Reading and Writing (27 Questions)

Question 1

Answer: B

(Explanation: “Reductive” means presenting something in an oversimplified way that misrepresents the complexity of the original. The passage emphasizes that griot performances encompass legal testimony, genealogical record-keeping, and diplomatic negotiation—functions far too varied and sophisticated for a single Western literary term. Labeling all of this “folklore” is fundamentally reductive. “Arbitrary” (A) means random or without reason, but the label “folklore” is not random—it is a specific (if oversimplified) category. “Indulgent” (C) means excessively lenient or self-gratifying, which is irrelevant here. “Prescient” (D) means having foresight, which contradicts the criticism being made.)

Question 2

Answer: C

(Explanation: “Capricious” means governed by whim rather than reason or logic. The sentence structure sets up a contrast: critics dismiss the designs as _____, while supporters argue the geometry is “carefully calibrated to serve its intended function.” The blank must therefore mean something like “whimsical” or “random”—the opposite of calibrated and functional. “Capricious” captures this precisely. “Meticulous” (A) means careful and precise, which is what supporters claim, not critics. “Derivative” (B) means copied from others, which the passage never suggests. “Austere” (D) means severely plain, the opposite of curvilinear designs.)

Question 3

Answer: A

(Explanation: The passage draws a contrast: sleep deprivation does not merely impair recall—it actively does something stronger to memory formation itself. “Precludes” means prevents from happening entirely, which represents a stronger effect than impairment and fits the escalation signaled by “not merely...actively.” “Augments” (B) means increases or enhances, which is the opposite of the intended meaning. “Expedites” (C) means speeds up, also the opposite. “Conflates” (D) means merges two things together, which doesn’t apply to disrupting a biological process.)

Question 4

Answer: D

(Explanation: “Entrench” means to establish something so firmly that change becomes very difficult. The passage sets up a contrast with “rather than ameliorate them” (ameliorate = improve or reduce), so the blank must mean something opposite to amelioration—making inequalities worse or more fixed. “Entrench” fits this precisely: the agreements deepen and solidify existing inequalities. “Illuminate” (A) means to shed light on, which describes understanding, not worsening. “Reconcile” (C) means to make compatible or resolve differences, the opposite of what is described. “Circumvent” (B) means to go around or avoid, which doesn’t capture the idea of deepening inequality.)

Question 5

Answer: A

*(Explanation: The passage follows a clear two-part structure. The first half identifies the traits—short reproductive cycle, many offspring, well-mapped genome—that make *Drosophila* ideal for laboratory experiments. The second half, introduced by the pivot word “yet,” explains how “these very qualities” may limit the generalizability of the findings. Choice A precisely captures this structure: useful traits, then a limitation arising from those same traits. Choice B is wrong because the passage does not present two competing views—it presents one perspective that pivots. Choice C is wrong because no misconception is identified or refuted. Choice D is wrong because the passage does not describe the history of the organism’s use or evaluate specific discoveries.)*

Question 6

Answer: B

*(Explanation: The passage builds a narrative arc: Brock's 1969 discovery attracted "modest attention," then the underlined sentence—describing Mullis's 1983 development of PCR using an enzyme from *T. aquaticus*—marks the turning point when the organism's importance "became clear." The sentence identifies the specific development (PCR) that revealed the broader significance of the earlier discovery. Choice A is wrong because the sentence is about what changed the perception, not about why it was initially insignificant. Choice C is wrong because no competing applications are contrasted. Choice D is wrong because no biographical context about Mullis's interests is provided.)*

Question 7

Answer: A

(Explanation: The text has three parts: (1) Scott's argument that the shift to agriculture was coerced by emerging states, (2) bioarchaeological evidence supporting this reappraisal, and (3) Ferraro's qualification that Scott may overstate the role of coercion. The overall purpose is to present a scholarly argument and then introduce a qualification—a common academic structure. Choice B is wrong because Tanaka and Okonkwo do not appear, and the text doesn't describe an "unrecognized issue." Choice C is wrong because no definitive resolution is claimed—the text acknowledges Ferraro's counterpoint. Choice D is wrong because Ferraro qualifies Scott rather than rejecting him outright, and the text never says economic historians have rejected anthropological conclusions.)

Question 8

Answer: B

(Explanation: Chomsky claims language acquisition is rapid because an innate grammar is "already in place at birth." Tomasello's research, as described in Text 2, directly counters this by showing that children's early utterances are "item-specific and formulaic, becoming more abstract only gradually"—a trajectory he calls "inconsistent with the sudden activation of an innate system." Tomasello would therefore contend that the gradual, pattern-driven nature of early language use suggests grammar is constructed through experience, not innately specified. Choice A is wrong because Tomasello doesn't challenge the speed of acquisition—he offers an alternative explanation

for it. Choice C is wrong because Text 2 challenges the existence of the innate framework, not just its relative importance. Choice D is wrong because Tomasello doesn't argue that cross-cultural variation undermines nativist claims.)

Question 9

Answer: A

(Explanation: The passage's central claim is that computational analysis reveals the boundary between "literary" and "popular" Victorian fiction was "far more permeable than traditional literary histories suggest." The passage explicitly states this does not diminish canonical authors but shows their works participated in a broader ecosystem. Choice B overstates the claim: the passage says canonical authors shared features with popular fiction, not that they were "primarily influenced" by sensational fiction. Choice C mischaracterizes the argument—the passage says findings "do not diminish the achievements of canonical authors," so calling them "less innovative" contradicts the text. Choice D makes a comparison with modern magazines that the passage never draws.)

Question 10

Answer: B

(Explanation: The ecologist's hypothesis is that precipitation, not elevation, primarily determines vegetation type. To support this, we need data showing that precipitation predicts vegetation better than elevation does. Nyeri has a lower elevation (1,759 m) than Nanyuki (1,947 m), yet Nyeri receives significantly more precipitation (988 mm vs. 680 mm) and supports denser vegetation (mixed forest vs. montane grassland). If elevation alone drove vegetation, Nyeri should have sparser growth—but it doesn't, because its higher precipitation supports denser vegetation. Choice A incorrectly treats elevation and precipitation as interchangeable. Choice C incorrectly suggests elevation drives precipitation. Choice D misstates the data—Nanyuki and Nyeri support different vegetation types.)

Question 11

Answer: B

(Explanation: The team concluded that eruptions show "pronounced clustering, with periods of intense activity separated by long quiescent intervals." To directly support this, we need evidence showing that tephra layers are grouped together in time with gaps be-

tween groups—exactly what choice B describes: “statistically significant groupings at intervals of approximately 40,000 to 80,000 years, with intervening gaps containing few or no layers.” Choice A (same volcanic source) is about origin, not frequency patterns. Choice C (record extends further back) addresses the extent of the record, not the distribution of eruptions within it. Choice D (better preservation of recent layers) could actually undermine the conclusion by suggesting gaps may be artifacts of poor preservation.)

Question 12

Answer: B

(Explanation: The hypothesis has two components: (1) at extreme depths with no ambient light, brighter lures attract prey without extra predation risk, and (2) at moderate depths with some ambient light, dimmer lures avoid detection by predators above. Choice B directly confirms both components: the brightest lures belong to species at the darkest depths, and moderate-depth species produce lures too dim for predators to see. Choice A (different chemical reactions) describes the mechanism of light production, not whether brightness is adapted to depth. Choice C (similar metabolic rates) rules out an alternative explanation but doesn’t positively confirm the depth-brightness relationship. Choice D (wide range of prey attraction) doesn’t address predation risk, which is central to the hypothesis.)

Question 13

Answer: B

(Explanation: The conclusion states that “the cognitive cost of language switching diminishes with bilingual proficiency,” measured by neural activation in executive control regions. Choice B directly supports this: participants in the top proficiency quartile showed a smaller difference in prefrontal cortex (an executive control region) activation between switching and non-switching trials than those in the bottom quartile—meaning higher proficiency correlates with lower cognitive cost. Choice A contradicts the conclusion (daily users showing slower times). Choice C describes auditory processing, not executive control regions. Choice D compares brain volume by age of acquisition, which is not relevant to the cognitive cost of switching.)

Question 14

Answer: C

(Explanation: The claim describes a conflict between moral duty and self-preservation. Choice C explicitly depicts both sides: “Margaret’s conscience told her to speak” represents moral duty, while “every nerve in her body urged her to stay silent, to let the moment pass safely by” represents the instinct for self-preservation. The two impulses are presented in direct opposition within a single sentence. Choice A shows emotional distress but not a conflict between two specific opposing forces. Choice B shows restlessness without identifying either duty or self-preservation. Choice D describes a decisive action already taken, not an ongoing internal conflict between competing impulses.)

Question 15

Answer: A

(Explanation: The prevailing model attributes bleaching primarily to thermal stress. Andersen, however, documented bleaching where “water temperatures remained within the historical range.” If bleaching occurs even when temperatures are normal, then thermal stress alone cannot fully account for the phenomenon—other factors must also contribute. This logically suggests that bleaching “may be driven by factors beyond thermal stress alone.” Choice B mischaracterizes the observation—Andersen’s work doesn’t question the role of symbiotic algae. Choice C introduces a claim about resistance to future bleaching that is not supported. Choice D contradicts Andersen’s own findings, which show that bleaching does occur in stable-temperature regions.)

Question 16

Answer: A

(Explanation: Latour argues that germ theory’s acceptance depended less on the strength of experimental evidence and more on Pasteur’s ability to recruit powerful allies—physicians, public health officials, and agricultural interests—who had their own pragmatic reasons for endorsing the framework. If this account is correct, acceptance was achieved through coalition-building and institutional support as much as through scientific persuasion, making it “a political and institutional achievement as much as a purely scientific one.” Choice B overstates Latour’s claim—he doesn’t say evidence is unnecessary, only that it was insufficient alone. Choice C incorrectly calls this an unrepeatable anomaly. Choice D makes a narrow comparison between Pasteur and Koch that the passage doesn’t support.)

Question 17**Answer: A**

(Explanation: The opening participial phrase “Engineered to withstand sustained temperatures exceeding 1,600 degrees Celsius” must modify the subject “the new generation of ceramic thermal shields.” The sentence then needs a main verb for this subject. Choice A, “are designed to protect,” provides a finite verb that correctly completes the predicate: “shields...are designed to protect crewed spacecraft during atmospheric reentry.” Choice B (“designing the protection of”) creates a participial fragment with no main verb. Choice C (“having been designed protecting”) stacks participles awkwardly without a finite verb. Choice D (“and are designed to protect”) creates a compound predicate with “unveiled,” but “unveiled” is part of a prepositional phrase modifying “shields,” not a coordinate verb.)

Question 18**Answer: B**

(Explanation: A colon is used to introduce an explanation, elaboration, or specification of what precedes it. Here, the first clause states that the excavation “upended a central assumption,” and the second clause specifies what that assumption was: “complex monumental construction...required the social organization that only settled agricultural communities could provide.” The colon connects these logically. A semicolon (A) separates two independent clauses of equal weight, but here the second clause explains the first. A comma (C) creates a comma splice between two independent clauses. “—and” (D) suggests an addition rather than an elaboration.)

Question 19**Answer: C**

(Explanation: In a “neither...nor” construction, the verb agrees with the nearer subject—in this case, “the implementation of new import tariffs on finished goods,” which is singular. The singular present-tense verb “is” correctly agrees with this subject. “Are” (A) is plural and does not agree with the singular nearer subject. “Were” (B) is plural past tense. “Have been” (D) is plural present perfect. All three violate the proximity agreement rule for “neither...nor” constructions.)

Question 20**Answer: B**

(Explanation: “By the time...published” establishes a past reference point (1980). The data collection began nearly two decades earlier and was completed before the publication. The past perfect “had spent” correctly indicates an action that was completed before another past event. “Has spent” (A) is present perfect, which links past actions to the present—inappropriate with the past reference point “published.” “Was spending” (C) is past continuous, which implies the action was still ongoing at the moment of publication rather than completed over the full period. “Would spend” (D) is conditional or future-in-the-past, suggesting the action had not yet begun.)

Question 21**Answer: A**

(Explanation: “Whose symphonies were most frequently performed during the 2023 Berlin Philharmonic season” is a restrictive (essential) relative clause—it identifies which specific composer is being discussed. Restrictive clauses are not set off by commas because the information is necessary for meaning. Choice A, “were” with no commas, is correct. Choice B places a comma before “were,” incorrectly separating the relative pronoun from its verb. Choice C places a comma after “were,” breaking the clause. Choice D encloses “were” in commas, making the clause nonrestrictive, which would incorrectly suggest the composer has already been identified.)

Question 22**Answer: D**

(Explanation: The sentence contains a three-item list of actions after “calls for”: “digitizing all archival photographs,” “_____ the humidity levels,” and “retraining staff.” Parallel structure requires all three items to take the same grammatical form. Since “digitizing” and “retraining” are gerunds (present participles used as nouns), the middle item must also be a gerund: “regulating.” “To regulate” (A) is an infinitive, breaking the parallel. “They will regulate” (C) introduces an independent clause, disrupting the list. “Regulation of” (B) is a noun phrase, not a gerund, and also breaks the parallel structure.)

Question 23**Answer: B**

(Explanation: The first sentence describes one engineering challenge: diverting the entire Colorado River through tunnels. The second sentence introduces a separate, additional engineering challenge: the concrete cooling problem. “Moreover” correctly signals that the second point adds to and reinforces the first—both illustrate the enormous engineering difficulties of dam construction. “Similarly” (A) implies the two challenges are alike in nature, but river diversion and concrete cooling are different types of problems. “In other words” (C) implies a restatement of the same idea, not a new challenge. “Nevertheless” (D) signals a contrast or concession, which is not the relationship between the two sentences.)

Question 24

Answer: C

(Explanation: The goal is to emphasize the contrast between conservation achievements and ongoing vulnerability. Choice C accomplishes this with the concessive clause “Although a conservation program has increased the kākāpō population from 51 individuals in 1995 to roughly 250 by 2024” (the achievement) directly followed by “the species remains critically endangered due to low genetic diversity and vulnerability to introduced predators” (the ongoing vulnerability). The word “although” explicitly creates the contrast. Choice A presents only the decline without mentioning conservation efforts or ongoing threats. Choice B presents only the achievement without addressing ongoing vulnerability. Choice D mentions both the program and endangered status but doesn’t articulate the contrast between success and remaining threats.)

Question 25

Answer: C

(Explanation: The goal is to explain why the Dead Sea supports some life (halophilic microorganisms) but not others (fish). Choice C directly addresses both sides: the 34% salinity “is lethal to fish and other macroscopic aquatic organisms,” yet halophilic archaea and bacteria “thrive in the water because they are adapted to extreme salt concentrations.” The causal reasoning—salinity kills some organisms while adapted organisms survive—precisely answers the question. Choice A provides facts about salinity but doesn’t explain why some life survives. Choice B describes location and elevation, which are irrelevant to the biological question. Choice D contains a factual error—it incorrectly states the Dead Sea supports fish, contradicting the notes.)

Question 26**Answer: C**

(Explanation: The goal is to draw a connection between the two projects. Choice C identifies the shared design principle—"geometric simplicity to create modern structures that complement their surrounding historic architecture"—and applies it to both projects by name (East Building, 1978; Louvre Pyramid, 1989). This explicitly connects the two works through their common architectural philosophy. Choice A discusses only the East Building. Choice B mentions both but only notes a chronological gap, not a design connection. Choice D discusses only the Louvre Pyramid as a landmark without linking it to the East Building or identifying a shared principle.)

Question 27**Answer: B**

(Explanation: The goal is to emphasize the trend in bleaching frequency and its implications for coral recovery. Choice B accomplishes all three elements: it identifies the trend ("mass bleaching events at increasingly short intervals"), names the cause ("ocean temperatures rise due to climate change"), and states the implication ("leaving less time between episodes for damaged coral to recover"). Choice A mentions multiple bleaching years but doesn't identify the shortening trend or discuss recovery implications. Choice C attributes bleaching to climate change but omits the trend in frequency and recovery implications. Choice D describes the reef's location and size, which is irrelevant to the trend and recovery question.)

SECTION 2, HARD MODULE 2: Math (22 Questions)**Question 1****Answer: C**

(Explanation: Set equal: $x^2 - 6x + 5 = 2x - 7 \rightarrow x^2 - 8x + 12 = 0 \rightarrow (x - 2)(x - 6) = 0 \rightarrow x = 2$ or $x = 6$. Two solutions: (2, -3) and (6, 5).)

Question 2**Answer: C**

(Explanation: Factor the numerator: $2x^2 + 9x + 9 = (2x + 3)(x + 3)$. Factor the denominator: $x^2 - 9 = (x - 3)(x + 3)$. Cancel $(x + 3)$: result

is $(2x + 3)/(x - 3)$.)

Question 3

Answer: C

(Explanation: If the positive integer roots are p and q , then $pq = 36$ and $p + q = k$. The ordered factor pairs of 36 where both factors are positive integers are (1, 36), (2, 18), (3, 12), (4, 9), and (6, 6). These give $k = 37, 20, 15, 13$, and 12, respectively—5 distinct values.)

Question 4

Answer: D

(Explanation: $g(4) = 4^2 + 1 = 17$. $f(g(4)) = f(17) = 3(17) - 2 = 51$.)

Question 5

Answer: 2

(Explanation: $243 = 3^5$. So $3^{(2x+1)} = 3^5 \rightarrow 2x + 1 = 5 \rightarrow 2x = 4 \rightarrow x = 2$.)

Question 6

Answer: A

(Explanation: Divide $x^3 - 2x^2 - 13x - 10$ by $(x + 1)$ using synthetic division with -1 : coefficients 1, -2 , -13 , -10 . Bring down 1. Multiply by -1 : -1 . Add to -2 : -3 . Multiply by -1 : 3. Add to -13 : -10 . Multiply by -1 : 10. Add to -10 : 0. Quotient: $x^2 - 3x - 10 = (x - 5)(x + 2)$. So $p(x) = (x + 1)(x - 5)(x + 2)$. Check: $(x + 1)(x - 5) = x^2 - 4x - 5$. Then $(x^2 - 4x - 5)(x + 2) = x^3 + 2x^2 - 4x^2 - 8x - 5x - 10 = x^3 - 2x^2 - 13x - 10$ ✓.)

Question 7

Answer: B

(Explanation: Complete the square: $(x^2 - 8x + 16) + (y^2 + 6y + 9) = 24 + 16 + 9 \rightarrow (x - 4)^2 + (y + 3)^2 = 49$. Radius = $\sqrt{49} = 7$.)

Question 8

Answer: A

(Explanation: $\sin^2\theta + \cos^2\theta = 1 \rightarrow \cos^2\theta = 1 - 25/169 = 144/169 \rightarrow \cos\theta = \pm 12/13$. In the second quadrant, cosine is negative: $\cos\theta = -12/13$.)

Question 9

Answer: D

(Explanation: Since $DE \parallel BC$, triangles ADE and ABC are similar. $AD/AB = DE/BC \rightarrow 6/(6 + 9) = 8/BC \rightarrow 6/15 = 8/BC \rightarrow BC = 8 \times 15/6 = 120/6 = 20$.)

Question 10

Answer: D

(Explanation: Set A: all identical, $SD = 0$. Set B: mean = 10, $SD \approx 1.41$. Set C: mean = 10, $SD \approx 5.66$. Set D: mean = 18, deviations are $-8, -8, -8, -8, 32$; variance = $(64+64+64+64+1024)/5 = 256$, $SD = 16$. Set D has the greatest standard deviation.)

Question 11

Answer: B

(Explanation: $2x^2 - 12x + 23 = 2(x^2 - 6x) + 23 = 2(x^2 - 6x + 9 - 9) + 23 = 2(x - 3)^2 - 18 + 23 = 2(x - 3)^2 + 5$. The minimum value is 5, occurring when $x = 3$.)

Question 12

Answer: C

(Explanation: Let $r = \text{roses}$, $l = \text{lilies}$. $r + l = 12$ and $3r + 5l = 44$. From the first: $l = 12 - r$. Substitute: $3r + 5(12 - r) = 44 \rightarrow 3r + 60 - 5r = 44 \rightarrow -2r = -16 \rightarrow r = 8$. Check: 8 roses + 4 lilies = 12 ✓. Cost: $24 + 20 = 44$ ✓.)

Question 13

Answer: 4

(Explanation: $(x^{(2/3)})^6 = x^{(6 \times 2/3)} = x^{(12/3)} = x^4$. So $n = 4$.)

Question 14**Answer: 8**

(Explanation: $a^2 - b^2 = (a - b)(a + b)$. So $40 = 5(a + b) \rightarrow a + b = 8$.)

Question 15**Answer: B**

(Explanation: $x^2 - 5x - 6 = (x - 6)(x + 1)$. The roots are $x = -1$ and $x = 6$. The parabola opens upward (positive leading coefficient), so the expression is negative between the roots: $-1 < x < 6$.)

Question 16**Answer: C**

(Explanation: Arc length $= r\theta = 10 \times 3\pi/5 = 30\pi/5 = 6\pi$ centimeters.)

Question 17**Answer: B**

(Explanation: Compound interest formula: $A = P(1 + r)^t = 2,000(1.06)^t$.)

Question 18**Answer: C**

(Explanation: Let $m =$ muffins, $c =$ cookies. $m + c = 50$ and $3m + 2c = 130$. From the first: $c = 50 - m$. Substitute: $3m + 2(50 - m) = 130 \rightarrow 3m + 100 - 2m = 130 \rightarrow m = 30$. Check: 30 muffins + 20 cookies $= 50$ ✓. Revenue: $90 + 40 = 130$ ✓.)

Question 19**Answer: D**

(Explanation: $(1/r) + (1/s) = (r + s)/(rs)$. By Vieta's formulas: $r + s = 15/3 = 5$, and $rs = 11/3$. So $(1/r) + (1/s) = 5/(11/3) = 15/11$.)

Question 20**Answer: B**

(Explanation: $V = (1/3)\pi r^2 h = (1/3)\pi(25)(12) = (1/3)(300\pi) = 100\pi$.)

Question 21**Answer: A**

(Explanation: Let the original price be 100. After 20% increase: 120. After 25% decrease: $120 \times 0.75 = 90$. Net change: $(90 - 100)/100 = -10\%$.)

Question 22**Answer: C**

(Explanation: Slope of line m : $(10 - 4)/(3 - 1) = 6/2 = 3$. Slope of perpendicular line n : $-1/3$. Using point-slope with $(6, 2)$: $y - 2 = -(1/3)(x - 6) \rightarrow y - 2 = -x/3 + 2 \rightarrow y = -x/3 + 4$. The y -intercept is 4.)